

ALI HASSAN

Fullstack Developer

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Summary

Fullstack Developer building end-to-end web applications with **React/Next.js** on the frontend and **Node.js/Express** on the backend using TypeScript. Experienced in **RESTful APIs**, **WebSockets**, **PostgreSQL/MongoDB**, and cloud tools like **AWS S3 and Docker**. Focused on clean architecture, secure auth (**JWT**, **OAuth2**), and reliable validation (**Zod**, **Yup**).

Education

Egyptian E-Learning University

Bachelor of Computer Science

Sep. 2022 – May 2026

Alexandria, Egypt

Certifications & Courses

Advanced Node.js: Node.js Internals, Redis, AWS S3, CI/CD

Udemy — Instructor: Stephen Grider

2025

Node.js: REST & GraphQL APIs, Cloud Deployment

Udemy — Instructors: Andrei Neagoie, Adam Odziemkowski

2025

NestJS: The Complete Developer's Guide

Udemy — Instructor: Stephen Grider

2026

Experience

Prime Academy | *React.js, Node.js*

Fullstack Developer

Feb 2025 – Present

Freelance

- Built a modern E-learning platform serving **800+ students** with video playback, interactive quizzes, real-time chat, and an admin dashboard.
- Optimized API response times from **2s down to 500ms** through query optimization and efficient middleware design.
- Optimized Vite build configuration, **reducing chunk size by 50%** to improve load performance and production bundle efficiency.
- Handled file uploads via **Cloudflare R2** with uppy uploader.
- Implemented secure JWT authentication and role-based access control.
- Used Drizzle ORM with MySQL to model and manage relational data efficiently.
- Architected a multi-stage, multi-container Docker Compose setup with horizontally scaled backend replicas and Nginx-served frontend, using health-check-gated dependencies and non-root runtime users for production readiness.

CloudCode Mentor | *Next.js, Express, TypeScript*

Fullstack Developer

Nov 2025 – Apr 2026

Freelance

- Built a two-sided mentorship platform supporting 1-on-1 and group sessions with Google Meet integration.
- Eliminated double-booking race conditions using Redis-based distributed locks across booking endpoints.
- Integrated LinkedIn SSO and People API to streamline onboarding by auto-populating user profile data.
- Automated mentor badge generation using Puppeteer and recorded session attendance via Google Reports API.
- Handled file storage with AWS S3, schema validation with Zod, and product analytics via PostHog.
- Built moderation features including user banning, appeal submissions, mentor reporting, and session ratings.

International Knowledge | *Node.js, Express.js, TypeScript, Socket.IO, MySQL, Redis*

Backend Developer

Oct 2025 – Jan 2026

Freelance

- Built a full-featured e-learning platform with live video sessions via **100ms**, real-time chat via **Socket.IO**, and direct browser-to-**Cloudflare R2** file uploads to keep the server lean.
- Integrated **Paymob** payment gateway with cron-based retry jobs to guarantee no payment operations were silently lost on server errors.
- Implemented multi-strategy authentication using **JWT** and **Google OAuth** via Passport.js with HTTP-only cookies and bcrypt password hashing.
- Delivered real-time notifications via **SSE** and enforced API protection with Redis-backed rate limiting and input sanitization.
- Applied **Zod** schema validation, Winston structured logging, and Helmet security headers across all API layers.

Projects

Health Bridge | Graduation Project | *Node.js, Express.js, PostgreSQL, TypeScript, Redis, AWS S3*

- Built a full-stack doctor-client booking platform with AI chatbot, supporting bilingual (Arabic & English) interfaces and role-based access control.
- Engineered a custom scheduling system with recurring weekly slots, date-specific availability, rolling booking windows, and minimum notice hour enforcement.
- Integrated Google Workspace APIs (Meet, Calendar) to auto-generate video session links and send synchronized email confirmations upon booking.
- Resolved double-booking race conditions using Redis distributed locking with atomic SET NX EX, ownership tracking, exponential backoff, and heartbeat renewal.
- Optimized booking response times from 2–3s to under 500ms by splitting transactions: critical booking logic is awaited while Google Meet creation and notifications run asynchronously with cron-based retry fallback.
- Implemented real-time AI chatbot streaming via SSE with per-connection identity tracking and unique stream IDs, eliminating multi-user session conflicts.
- Offloaded patient prescription uploads to AWS S3 to prevent server I/O bottlenecks and maintain API responsiveness under concurrent load.

StockFlow | *NestJS, RabbitMQ, PostgreSQL, Redis, React*

- Built a microservices-based warehouse management system with an API gateway and four independent services communicating exclusively through RabbitMQ.
- Implemented global exception filters for consistent error response shapes across all services, and structured logging with Winston across the gateway and all services for centralized observability.
- Validated JWT authentication at the API gateway by querying the Auth service over RabbitMQ on every incoming request.
- Implemented background stock processing with BullMQ across the Orders and Inventory services, backed by Redis.
- Built a React and shadcn/ui dashboard for managing inventory, inbound/outbound orders, and payments.

Blogy | *Next.js, PostgreSQL, Redis*

- Developed a Markdown-powered blogging platform with live preview and chunked image uploads via FilePond.
- Integrated JWT-based authentication, role-based authorization, and Zod validation.
- Employed Redis for request rate limiting to mitigate abuse.
- Wrote unit and integration tests using Jest and automated CI/CD via GitHub Actions.

Soft Skills

Communication

Team Collaboration

Time Management

Problem Solving

Adaptability

Critical Thinking

Self-motivation

Integrity

Technical Skills

Languages: JavaScript, TypeScript

Frontend Technologies: React.js, Next.js, Tailwind CSS, ShadCN, Redux, Zustand, React Query

Backend Technologies: Node.js, Express.js, Nest.js, Next.js (API Routes), WebSockets, Server-Sent Events (SSE), GraphQL, Redis

ORMs & Databases: Prisma, Sequelize, Drizzle, TypeORM, PostgreSQL, MongoDB, MySQL, SQLite

Authentication & Security: JWT, OAuth2, HTTP-only cookies, Zod, Yup

Testing & Automation: Jest, Puppeteer

Developer Tools: VS Code, Git, Postman, Bun, Cloudflare Workers, Docker, PM2

Cloud & Deployment: Vercel, AWS (S3), Nginx

Languages

Arabic: Native

English: fluent